NORMALIZATION OF A FINITE TYPE SEPARATED MORPHISM OF ALGEBRAIC SPACES IS AN ISOMORPHISM OVER AN OPEN SUBSPACE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7611-7707, immediately following the statement at 7592-7609.

Editorial note. Seven cross-chapter pointers (Morphisms of Spaces, Descent on Spaces) plus two same-chapter lemmas are recorded in assembly.external_references.

Editorial note. Same-name pair note: this is the more general (algebraic spaces) side; the schemes side is a separate statement in the other chapter and is not counted with this card.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 7632 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Etale splitting off of the quasi-finite part. chapter lemma lemma-etale-splits-off-quasi-finite-part, cited at proof line 7694. Produces, etale locally over a point, a decomposition into a finite part and a part whose normalization is followed.
- Localization of the smooth/normalization comparison. chapter lemma lemma-normalization-smooth-localization, cited in the proof. Lets the normalization be computed after localizing, which is what the local step needs.
- Normalization commutes with disjoint unions and with integral morphisms. Morphisms of Spaces, lemmas spaces-morphisms-lemma-normalization-in-disjoint-union and spaces-morphisms-lemma-normalization-in-integral, cited at proof lines 7697-7701 onward. Identifies the normalization of Y in X = V coproduct W with V coproduct W', which yields the open subspace U'.
- Residue field extension realised by a closed point of a fibre. Morphisms of Spaces, lemmas spaces-morphisms-lemma-algebraic-residue-field-extension-closed-point-fibre and spaces-morphisms-lemma-locally-quasi-finite-fibres, cited in the proof. Supplies the point over which the etale-local splitting is performed.

Source reference. Original lemma 082H: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7472–7472. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 082F: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6141–6141. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TE: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06RT: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 041Y: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

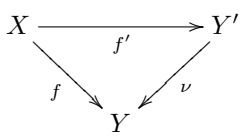
Source reference. Original reference 0414: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NW: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0824: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0825: focused-seeds/geometry/S9-spaces-finite-type-separated/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is of finite type and separated. Let Y' be the normalization of Y in X. Picture:*



Then there exists an open subspace $U' \subset Y'$ such that

(1) *$(f')^{-1}(U') \to U'$ is an isomorphism, and*
(2) *$(f')^{-1}(U') \subset X$ is the set of points at which f is quasi-finite.*

Proof. By Morphisms of Spaces, Lemma 04NW there is an open subspace $U \subset X$ corresponding to the points of $|X|$ where f is quasi-finite. We have to prove

(a) the image of $|U| \to |Y'|$ is $|U'|$ for some open subspace U' of Y',
(b) $U = f^{-1}(U')$, and
(c) $U \to U'$ is an isomorphism.

Since formation of U commutes with arbitrary base change (Morphisms of Spaces, Lemma 04NW), since formation of the normalization Y' commutes with smooth base change (Lemma 082F), since étale morphisms are open, and since "being an isomorphism" is fpqc local on the base (Descent on Spaces, Lemma 041Y), it suffices to prove (a), (b), (c) étale locally on Y (some details omitted). Thus we may assume Y is an affine scheme. This implies that Y' is an (affine) scheme as well.

Let $x \in |U|$. Claim: there exists an open neighbourhood $f'(x) \in V \subset Y'$ such that $(f')^{-1}V \to V$ is an isomorphism. We first prove the claim implies the lemma. Namely, then $(f')^{-1}V \cong V$ is a scheme (as an open of Y'), locally of finite type over Y (as an open subspace of X), and for $v \in V$ the residue field extension $\kappa(v)/\kappa(\nu(v))$ is algebraic (as $V \subset Y'$ and Y' is integral over Y). Hence the fibres of $V \to Y$ are discrete (Morphisms, Lemma 01TE) and $(f')^{-1}V \to Y$ is locally quasi-finite (Morphisms, Lemma 06RT). This implies $(f')^{-1}V \subset U$ and $V \subset U'$. Since x was arbitrary we see that (a), (b), and (c) are true.

Let $y = f(x) \in |Y|$. Let $(T, t) \to (Y, y)$ be an étale morphism of pointed schemes. Denote by a subscript $_T$ the base change to T. Let $z \in X_T$ be a point in the fibre X_t lying over x. Note that $U_T \subset X_T$ is the set of points where f_T is quasi-finite, see Morphisms of Spaces, Lemma 04NW. Note that

$$X_T \xrightarrow{f'_T} Y'_T \xrightarrow{\nu_T} T$$

is the normalization of T in X_T, see Lemma 082F. Suppose that the claim holds for $z \in U_T \subset X_T \to Y'_T \to T$, i.e., suppose that we can find an open neighbourhood $f'_T(z) \in V' \subset Y'_T$ such that $(f'_T)^{-1}V' \to V'$ is an isomorphism. The morphism $Y'_T \to Y'$ is étale hence the image $V \subset Y'$ of V' is open. Observe that $f'(x) \in V$ as $f'_T(z) \in V'$. Observe that

$$\begin{array}{ccc} (f'_T)^{-1}V' & \longrightarrow & (f')^{-1}(V) \\ \downarrow & & \downarrow \\ V' & \longrightarrow & V \end{array}$$

is a fibre square (as $Y'_T \times_{Y'} X = X_T$). Since the left vertical arrow is an isomorphism and $\{V' \to V\}$ is a étale covering, we conclude that the right vertical arrow is an isomorphism by Descent on Spaces, Lemma 041Y. In other words, the claim holds for $x \in U \subset X \to Y' \to Y$.

By the result of the previous paragraph to prove the claim for $x \in |U|$, we may replace Y by an étale neighbourhood T of $y = f(x)$ and x by any point lying over x in $T \times_Y X$. Thus we may assume there is a decomposition

$$X = V \amalg W$$

into open and closed subspaces where $V \to Y$ is finite and $x \in V$, see Lemma 082H. Since X is a disjoint union of V and W over Y and since $V \to Y$ is finite we see that the normalization of Y in X is the morphism

$$X = V \amalg W \longrightarrow V \amalg W' \longrightarrow S$$

where W' is the normalization of Y in W, see Morphisms of Spaces, Lemmas 0824, 0414, and 0825. The claim follows and we win. □